

A stabilized LSMPS for three-dimensional complex free-surface flow with moving wall

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ABSTRACT

This study proposes a stabilized LSMPS method for simulating free surface flows with moving wall. In this stabilized LSMPS method, a novel wall velocity model is proposed for velocity divergence calculation and velocity interpolation. The proposed method also combines type-A and type-B LSMPS to enhance both accuracy and stability. Type-A LSMPS is applied to internal particles, while type-B scheme is used for free-surface and sub-free-surface particles. Additionally, type-B LSMPS is employed to interpolate the velocity after particle shifting. The stability of both type-A and type-B schemes is enhanced by the incorporation of positive values into the diagonal element of the least-squares matrix. To verify the developed LSMPS method, several three-dimensional numerical simulations are conducted.

1. Introduction

Free-surface flows with moving wall, characterized by dynamic surface deformation and moving wall boundaries have been extensively encountered across a wide range of industrial applications. Over the past few decades, extensive research has been conducted to numerically investigate the dynamics of these flows. However, accurately simulating fluids with complex free surfaces and moving wall boundaries continues to pose significant challenges.

In recent years, meshless moving particle methods have demonstrated significant potential in addressing free-surface flow challenges. Especially in the analysis of free-surface flow with moving wall, meshless moving particles methods have demonstrated its potential in handling complex geometries. These methods, which eliminate the need for grid generation, leverage their Lagrangian nature to effectively simulate free-surface flows. Among these, the smoothed particle hydrodynamics (SPH) method [1] and the moving particle semi-implicit (MPS) method [2] are recognized as classical approaches. SPH and MPS have been extensively applied in engineering analyses, particularly in the fields of mechanical, ocean and nuclear engineering [3–8].

The accuracy and dependability of the MPS and SPH methods are significantly reduced when particle distribution becomes irregular and discretization errors increase substantially. In the realm of meshless moving particle methods, the advancement of high-order gradient and Laplacian models is particularly crucial. Several notable techniques have been developed to enhance the precision of particle methods, including the corrective matrix method [9–13], the least squares method [14,15], the Lagrangian-Eulerian stabilized collocation method [16,17] and the pairwise-relaxing method [18,19]. Among these, the Least Squares Moving Particle Semi-implicit (LSMPS) method stands out for its ability to discretize differential operators with arbitrary accuracy.

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Initially, the application of LSMPS is only limited to internal flows [20,21]. Matsunaga and Koshizuka [22] developed a stabilized LSMPS for complex free-surface flow simulation. In their method, the free surface was fitted and regularized least squares were adopted. Duan et al. [23] investigates the error accumulation due to biased particle distribution and developed a novel free-surface identification method in LSMPS when simulating free surface flow. The stabilized LSMPS is extended to compact LSMPS method for better stability and accuracy [24] when simulating incompressible free-surface flow. To address the volume non-conservation issue of LSMPS method when simulating incompressible free-surface flow, Liu et al. [25,26] developed the volume conservation term for both single-resolution and multi-resolution LSMPS.

Although the LSMPS has been studied comprehensively in the area of free-surface flow, most of the study is two-dimensional study and application of LSMPS for three-dimensional complex free surface flow with moving wall has been rarely mentioned. LSMPS needs to calculate the inversion of the moment matrix. This calculation brings into large instability issue for the LSMPS simulating free-surface flow. One needs to carefully choose the order and type of LSMPS scheme and using regularization term to stabilize the calculation. In addition, the meaning of the regularization term has not been well clarified.

The present study investigates the accuracy and stability of LSMPS in three-dimensional space. A novel approach was proposed, entailing the integration of the type-A and type-B schemes to ensure the stability of the LSMPS calculation. Additionally, the wall boundary condition has been enhanced to facilitate the effective implementation of LSMPS in complex three-dimensional calculations with moving wall.

The rest of the paper is organized as follows. In section2, we outline the LSMPS method, providing the details of type-A LSMPS and type-B LSMPS. In section 3, the stabilized LSMPS developed in this study is presented, including the boundary condition and stability treatment. Section 4 presents the numerical results for a number of numerical tests. A summary of the key findings is concluded in section 5.

2. LSMPS method

The LSMPS method solves the following governing equations for incompressible flows:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho}\nabla p + \frac{\mu}{\rho}\nabla^2\mathbf{u} + \mathbf{f}, \quad (2)$$

where $\mathbf{u}$ is the velocity, t is time, ρ is density, μ is viscosity, p is pressure and $\mathbf{f}$ represents other forces, such as gravity force.

2.1. Discretization method

This study employs LSMPS developed by Tamai and Koshizuka [15] to discretize the governing equations. LSMPS includes type A and type B scheme. In the type-A scheme, the Taylor series expansion of a scalar variable around a specific point with respect to a neighboring particle is expressed as:

$$\phi_j = \phi_i + \sum_{m=1}^p \frac{1}{m!} (\mathbf{r}_{ij} \cdot \nabla)^m \phi|_{r_i} + o(|\mathbf{r}_{ij}|^{p+1}) \quad (3)$$

Ignoring the higher-order terms, one can obtain the following equation:

$$\phi_j - \phi_i = \sum_{m=1}^p \frac{r_{ij}^m}{m!} \left(\frac{\mathbf{r}_{ij}}{r_s} \cdot \nabla \right)^m \phi|_{r_i} = \mathbf{p}_{ij} \cdot \left(\mathbf{H}_{r_s}^{-1} \Lambda \phi|_{r_i} \right) \quad (4)$$

where $\phi_i = \phi(\mathbf{r}_i)$, $\phi_j = \phi(\mathbf{r}_j)$, $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, r_s denotes a scaling parameter, p is a positive integer, which indicates the order of approximation, $\mathbf{p}_{ij}$ denotes a polynomial basis vector, $\mathbf{H}_{r_s}^{-1}$ represents a scaling matrix and Λ indicates a differential operator vector.

For the second order scheme ($p = 2$) in three dimensions, these functions are written as:

$$\mathbf{p}_{ij} = \mathbf{p} \left(\frac{\mathbf{r}_{ij}}{r_s} \right) = \left[\frac{x_{ij}}{r_s}, \frac{y_{ij}}{r_s}, \frac{z_{ij}}{r_s}, \frac{x_{ij}^2}{r_s^2}, \frac{y_{ij}^2}{r_s^2}, \frac{z_{ij}^2}{r_s^2}, \frac{x_{ij}y_{ij}}{r_s^2}, \frac{x_{ij}z_{ij}}{r_s^2}, \frac{y_{ij}z_{ij}}{r_s^2} \right] \quad (5)$$

$$\mathbf{H}_{r_s} = \text{diag} \left(\frac{1}{r_s}, \frac{1}{r_s}, \frac{1}{r_s}, \frac{2}{r_s^2}, \frac{2}{r_s^2}, \frac{2}{r_s^2}, \frac{1}{r_s^2}, \frac{1}{r_s^2}, \frac{1}{r_s^2} \right) \quad (6)$$

$$\Lambda = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}, \frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial z^2}, \frac{\partial^2}{\partial x \partial y}, \frac{\partial^2}{\partial x \partial z}, \frac{\partial^2}{\partial y \partial z} \right]^T \quad (7)$$

By incorporating all neighboring particles and nodes denoted as j in Eq. (4), the objective function J for the weighted least squares problem is formulated as:

$$J(\mathbf{X}_i) = \sum_{j \neq i} w_{ij} (\mathbf{p}_{ij} \cdot \mathbf{X}_i - \phi_j + \phi_i)^2 \quad (8)$$

with

$$\mathbf{X}_i = \mathbf{H}_{r_s}^{-1} \Lambda \phi|_{r_i} \quad (9)$$

The following weight function w is extensively utilized in the LSMPS method:

$$w(|\mathbf{r}_{ij}|) = \begin{cases} \left(1 - \frac{|\mathbf{r}_{ij}|}{r_e}\right)^2, & |\mathbf{r}_{ij}| < r_e \\ 0, & |\mathbf{r}_{ij}| \geq r_e \end{cases} \quad (10)$$

where r_e denotes the effective radius. In this study, r_e is set to $2.4l_0$ for three-dimensional LSMPS calculation.

Minimization of J leads to the normal equations:

$$\mathbf{M}_i \mathbf{X}_i = \mathbf{b}_i \quad (11)$$

with

$$\mathbf{M}_i = \sum_{j \neq i} w_{ij} \mathbf{p}_{ij} \otimes \mathbf{p}_{ij} \quad (12)$$

$$\mathbf{b}_i = \sum_{j \neq i} w_{ij} \mathbf{p}_{ij}^T (\phi_j - \phi_i) \quad (13)$$

If the moment matrix $\mathbf{M}_i$ is nonsingular, a unique solution can be derived, and the spatial derivatives are determined as follows:

$$\partial \phi|_{r=r_i} = \mathbf{H}_{r_s} \mathbf{M}_i^{-1} \mathbf{b}_i \quad (14)$$

Finally, gradient, divergence and Laplacian model in type-A LSMPS can be outlined as

$$\langle \nabla \phi \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\phi_j - \phi_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \\ \mathbf{L}_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (15)$$

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j - \mathbf{u}_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \\ \mathbf{L}_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (16)$$

$$\langle \nabla^2 \phi \rangle_i = \sum_{j \neq i} \left\{ 2 \frac{w_{ij}}{r_s^2} (\phi_j - \phi_i) [\mathbf{L}_4 + \mathbf{L}_5 + \mathbf{L}_6] \cdot \mathbf{p}_{ij}^T \right\} \quad (17)$$

where $\mathbf{L}_k$ represents the k -th row of the matrix $\mathbf{M}_i^{-1}$.

In type-B scheme, $\phi(\mathbf{r}_i)$ is considered as an unknown value. Thus, a Taylor series expansion is expressed as:

$$\phi_j = \sum_{m=0}^p \frac{1}{m!} (\mathbf{r}_{ij} \cdot \nabla)^m \phi|_{r_i} + o(|\mathbf{r}_{ij}|^{p+1}) \quad (18)$$

By ignoring higher-order terms, one can obtain the following equations:

$$\phi_j = \sum_{m=0}^p \frac{r_s^m}{m!} \left(\frac{\mathbf{r}_{ij}}{r_s} \cdot \nabla \right)^m \phi|_{r_i} = \tilde{\mathbf{p}}_{ij} \cdot \left(\tilde{\mathbf{H}}_{r_s}^{-1} \tilde{\Lambda} \phi|_{r_i} \right) \quad (19)$$

where

$$\tilde{\Lambda} = \begin{bmatrix} 1 \\ \Lambda \end{bmatrix} \quad (20)$$

$$\tilde{\mathbf{H}}_{r_s} = \begin{bmatrix} 10 \\ 0\mathbf{H}_{r_s} \end{bmatrix} \quad (21)$$

$$\tilde{\mathbf{p}}_{ij} = \tilde{\mathbf{p}} \left(\frac{\mathbf{r}_{ij}}{r_s} \right) = \begin{bmatrix} 1 & \mathbf{p}_{ij} \end{bmatrix} \quad (22)$$

For the second order scheme ($p = 2$) in three dimensions, these functions are written as:

$$\tilde{\Lambda} = \left[1, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}, \frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial z^2}, \frac{\partial^2}{\partial x \partial y}, \frac{\partial^2}{\partial y \partial z}, \frac{\partial^2}{\partial x \partial z} \right]^T \quad (23)$$

$$\tilde{\mathbf{H}}_s = \text{diag}\left(1, \frac{1}{r_s}, \frac{1}{r_s}, \frac{1}{r_s}, \frac{2}{r_s^2}, \frac{2}{r_s^2}, \frac{2}{r_s^2}, \frac{1}{r_s^2}, \frac{1}{r_s^2}, \frac{1}{r_s^2}\right) \quad (24)$$

$$\tilde{\mathbf{p}}_{ij} = \tilde{\mathbf{p}}\left(\frac{\mathbf{r}_{ij}}{r_s}\right) = \left[1, \frac{x_{ij}}{r_s}, \frac{y_{ij}}{r_s}, \frac{z_{ij}}{r_s}, \frac{x_{ij}^2}{r_s^2}, \frac{y_{ij}^2}{r_s^2}, \frac{z_{ij}^2}{r_s^2}, \frac{x_{ij}y_{ij}}{r_s^2}, \frac{y_{ij}z_{ij}}{r_s^2}, \frac{x_{ij}z_{ij}}{r_s^2}\right] \quad (25)$$

Similarly, the spatial derivatives in the type-B scheme are obtained as:

$$\tilde{\Lambda}\phi_{r=r_i} = \tilde{\mathbf{H}}_s \tilde{\mathbf{M}}_i^{-1} \tilde{\mathbf{b}}_i \quad (26)$$

with

$$\tilde{\mathbf{M}}_i = \sum_j w_{ij} \tilde{\mathbf{p}}_{ij} \otimes \tilde{\mathbf{p}}_{ij} \quad (27)$$

$$\tilde{\mathbf{b}}_i = \sum_j w_{ij} \tilde{\mathbf{p}}_{ij}^T \phi_j \quad (28)$$

2.2. Time marching

The time-marching approach in the LSMPS method employs a semi-implicit scheme, where the viscous term and external forces are computed explicitly in the initial step:

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left[\frac{\mu}{\rho} \nabla^2 \mathbf{u}^n + \mathbf{f}^n \right] \quad (29)$$

where Δt is the time step size and $\mathbf{u}^*$ is the intermediate velocity.

Then the pressure Poisson equation can be derived as:

$$\nabla^2 p^{n+1} = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*, \quad (30)$$

After solving the pressure field from the above PPE, velocity is updated by considering the pressure gradient force:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p^{n+1} \quad (31)$$

Finally, particle position is updated:

$$\mathbf{r}^{n+1} = \mathbf{r}^n + \Delta t \cdot \mathbf{u}^{n+1}. \quad (32)$$

Usually, particle shifting scheme is adopted. Assuming the shifting vector is $\delta \mathbf{r}$, particle distribution is adjusted by

$$\tilde{\mathbf{r}}^{n+1} = \mathbf{r}^{n+1} + \delta \mathbf{r} \quad (33)$$

To ensure the accuracy of the overall calculation, velocity field is updated based on Taylor-series consistent interpolation:

$$\tilde{\mathbf{u}}^{n+1} = \mathbf{u}^{n+1} + \delta \mathbf{r} \cdot \nabla \mathbf{u}^{n+1} \quad (34)$$

The time step Δt is calculated based on the CFL condition:

$$\Delta t = 0.2 \frac{l_0}{u_{\max}}. \quad (35)$$

Here $u_{\max}$ is the maximum velocity of fluid particles.

3. Stabilized LSMPS

3.1. Boundary condition

At free surface, the following stress balance equation is imposed as the Dirichlet boundary condition for free surface particle i

$$P_{i,fs} = \mu \mathbf{n}_{i,fs}^T [\nabla \mathbf{u}_i + (\nabla \mathbf{u}_i)^T] \mathbf{n}_{i,fs} - \mathbf{n}_{i,fs}^T \cdot \mathbf{f}_{i,st} \quad (36)$$

where $P_{i,fs}$ is the pressure of free surface particle i , $\mathbf{n}_{i,fs}$ is the unit norm vector of the free surface and $\mathbf{f}_{i,st}$ is the surface tension force. The pressure value was directly assigned to free surface particles, and this pressure serves as the Dirichlet boundary condition in the solution of the PPE.

To detect free-surface particles more accurately, the geometry-based detection method developed by Shibata et al. [27] is adopted. In this geometry-based model, light source is defined at the center of the detection particle and virtual screen is defined surrounding

the detection particle. After calculating all the neighbor particles and counting the number of shaded segments, the illuminated fraction can be calculated used to identify free-surface particles.

For the wall boundary, the following Neumann boundary condition is imposed.

$$\frac{\partial p}{\partial \mathbf{n}}|^{n+1} = -\frac{\Delta t}{\rho} \mathbf{n}_{\text{wall}}^{n+1 T} \cdot (\mathbf{u}_{\text{wall}}^{n+1} - \mathbf{u}_{\text{wall}}^n - \Delta t \mathbf{g}) \quad (37)$$

Following [21], the normal gradient of physical quantity at the wall particle is approximated as

$$\frac{\partial \phi}{\partial \mathbf{n}}|_{r_j} = \mathbf{n}_j^T \sum_{m=0}^{p-1} \frac{r_s^m}{m!} \left(\frac{\mathbf{r}_{ij}}{r_s} \cdot \nabla \right)^m \nabla \phi|_{r_i} + o(|\mathbf{r}_{ij}|^p) \quad (38)$$

By ignoring the higher-order terms, Eq. (38) can be rewritten as:

$$r_s \frac{\partial \phi}{\partial \mathbf{n}}|_{r=r_j} = \mathbf{q}_{ij} \cdot (\mathbf{H}_{r_s}^{-1} \Lambda \phi|_{r_i}) \quad (39)$$

where $\mathbf{q}_{ij} = \frac{\partial}{\partial \mathbf{n}}(r_s \mathbf{p}_{ij})$ indicates a modified polynomial basis vector. Specifically, $\mathbf{q}_{ij}$ for $p = 2$ is given as:

$$\mathbf{q}_{ij} = \left[n_x, n_y, n_z, 2n_x \frac{x_{ij}}{r_s}, 2n_y \frac{y_{ij}}{r_s}, 2n_z \frac{z_{ij}}{r_s}, n_x \frac{y_{ij}}{r_s} + n_y \frac{x_{ij}}{r_s}, n_y \frac{z_{ij}}{r_s} + n_z \frac{y_{ij}}{r_s}, n_x \frac{z_{ij}}{r_s} + n_z \frac{x_{ij}}{r_s} \right] \quad (40)$$

The velocity impermeable boundary condition reads as

$$\mathbf{n}_{\text{wall}}^{n+1 T} \cdot \mathbf{u}^{n+1} = \mathbf{n}_{\text{wall}}^{n+1 T} \cdot \mathbf{u}_{\text{wall}}^{n+1} \quad (41)$$

After the explicit time step, the provisional velocity $\mathbf{u}^*$ on a wall particle is given by

$$\mathbf{n}_{\text{wall}}^{n+1 T} \cdot \mathbf{u}^* = \mathbf{n}_{\text{wall}}^{n+1 T} \cdot (\mathbf{u}_{\text{wall}}^n + \Delta t \mathbf{g}) \quad (42)$$

By taking the inner product of Taylor-series expansion of provisional velocity with the unit normal vector,

$$\mathbf{n}_j^T \cdot \mathbf{u}_j^* = \mathbf{n}_j^T \cdot \mathbf{u}_i^* + \mathbf{n}_j^T \cdot \sum_{m=1}^p \frac{r_s^m}{m!} \left(\frac{\mathbf{r}_{ij}}{r_s} \cdot \nabla \right)^m \mathbf{u}^*|_{r_i} + o(|\mathbf{r}_{ij}|^{p+1}) \quad (43)$$

By ignoring higher-order terms, the following equations for velocity wall boundary can be obtained

$$\begin{bmatrix} n_x \mathbf{p}_{ij} \\ n_y \mathbf{p}_{ij} \\ n_z \mathbf{p}_{ij} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{H}_{r_s}^{-1} \Lambda \mathbf{u}_i^* \\ \mathbf{H}_{r_s}^{-1} \Lambda \mathbf{v}_i^* \\ \mathbf{H}_{r_s}^{-1} \Lambda \mathbf{w}_i^* \end{bmatrix} \simeq \mathbf{n}_j^T \cdot (\mathbf{u}_{\text{wall}}^n + \Delta t \mathbf{g} - \mathbf{u}_i^*) \quad (44)$$

The combination of the boundary condition and the velocity divergence enables the calculation of velocity derivatives. However, this calculation involves the inversion of a 27th-order matrix in three-dimensional space. When the number of neighbor particles cannot be guaranteed, the 27th-order matrix may be ill-conditioned. To circumvent the necessity for the calculation of the 27th-order matrix, we propose the following model to approximate the velocity boundary condition.

$$\tilde{\mathbf{u}}_j^* = \mathbf{u}_i^* - (\mathbf{n}_{\text{wall}}^{n+1 T} \cdot \mathbf{u}_i^*) \mathbf{n}_{\text{wall}}^{n+1} + (\mathbf{n}_{\text{wall}}^{n+1 T} \cdot (\mathbf{u}_{\text{wall}}^n + \Delta t \mathbf{g})) \mathbf{n}_{\text{wall}}^{n+1} \quad (45)$$

When calculating the divergence of the provisional velocity of PPE, provisional velocity of a neighbor wall particle is replaced by $\tilde{\mathbf{u}}_j^*$. For instance, the divergence of the provisional velocity calculated by type-A LSMPS is

$$\langle \nabla \cdot \mathbf{u}^* \rangle_i = \sum_{j \neq i, j \in \text{fluid}} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j^* - \mathbf{u}_i^*) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \\ \mathbf{L}_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} + \sum_{j \neq i, j \in \text{wall}} \left\{ \frac{w_{ij}}{r_s} (\tilde{\mathbf{u}}_j^* - \mathbf{u}_i^*) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \\ \mathbf{L}_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (46)$$

Utilizing the aforementioned model, the difference between fluid velocity and provisional velocity on a wall particle in the tangential direction is disregarded, while the provisional velocity on a wall particle along the normal direction remains constant. The above velocity is directly incorporated into the type-A and type-B models. Inversion of the 27th-order matrix is avoided.

3.2. LSMPS mixing type-A and type-B

The stability of the LSMPS is of paramount importance, particularly in the vicinity of the free surface. In their research, several scholars have proposed the utilization of classical MPS in this area to enhance stability, albeit at the expense of accuracy. In this study, we propose a stable methodology involving the integration of type-A and type-B LSMPS schemes.

For the viscosity term and pressure Laplacian of PPE, type-A ($p = 2$) is used. Pressure of free surface particles is determined by the stress balance equation in Eq. (36) and used as Dirichlet boundary condition in the pressure Laplacian.

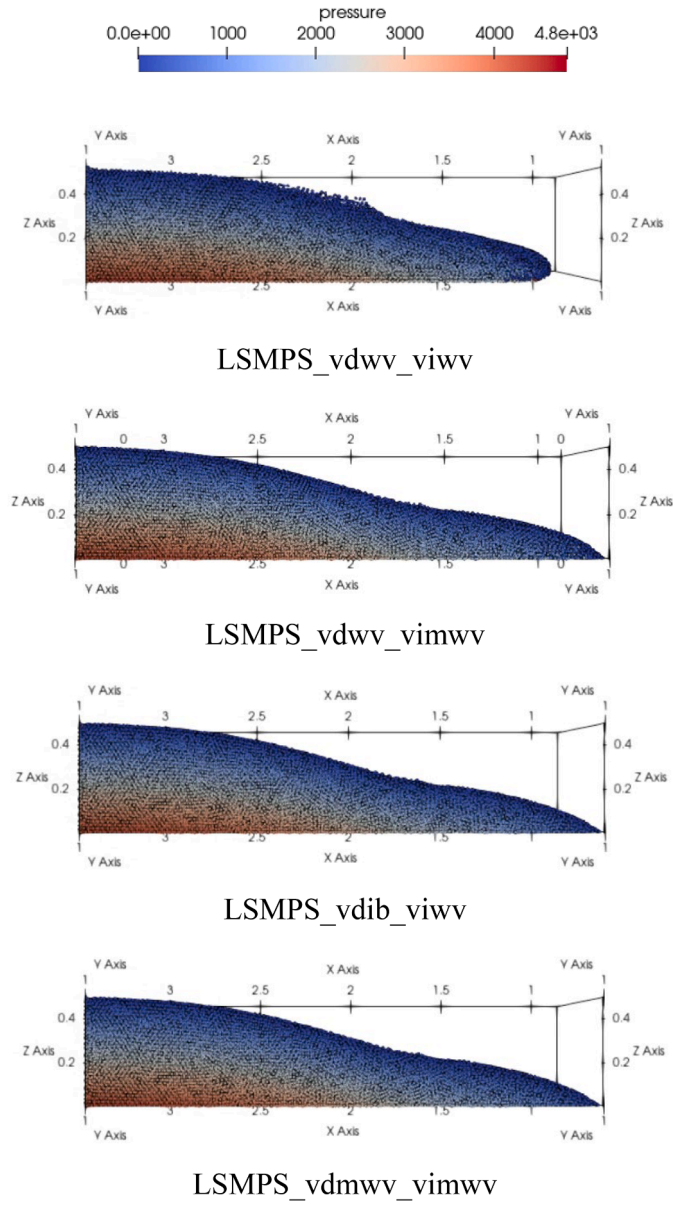


Fig. 1. Particle distribution together with pressure field of the dam at time = 0.4 s (side view).

$$\begin{aligned} \langle \nabla^2 p \rangle_i = & \sum_{j \neq i, j \in \text{freesurface}} \left\{ 2 \frac{w_{ij}}{r_s^2} (p_j - p_i) [\mathbf{L}_4 + \mathbf{L}_5 + \mathbf{L}_6] \cdot \mathbf{p}_{ij}^T \right\} \\ & + \sum_{j \neq i, j \in \text{freesurface}} \left\{ 2 \frac{w_{ij}}{r_s^2} (p_{jfs} - p_i) [\mathbf{L}_4 + \mathbf{L}_5 + \mathbf{L}_6] \cdot \mathbf{p}_{ij}^T \right\} \end{aligned} \quad (47)$$

In the viscosity model, non-slip boundary condition is used in the present study by setting

$$\mathbf{u}_j = \mathbf{u}_{\text{wall}} \quad j \in \text{wallparticle} \quad (48)$$

Velocity of the wall particle is set to the wall velocity and used in viscosity term calculation:

$$\langle \nabla^2 \mathbf{u} \rangle_i = \sum_{j \neq i} \left\{ 2 \frac{w_{ij}}{r_s^2} (\mathbf{u}_j - \mathbf{u}_i) [\mathbf{L}_4 + \mathbf{L}_5 + \mathbf{L}_6] \cdot \mathbf{p}_{ij}^T \right\} \quad (49)$$

For the pressure gradient and velocity divergence, type-A ($p = 2$) is used for internal particles:

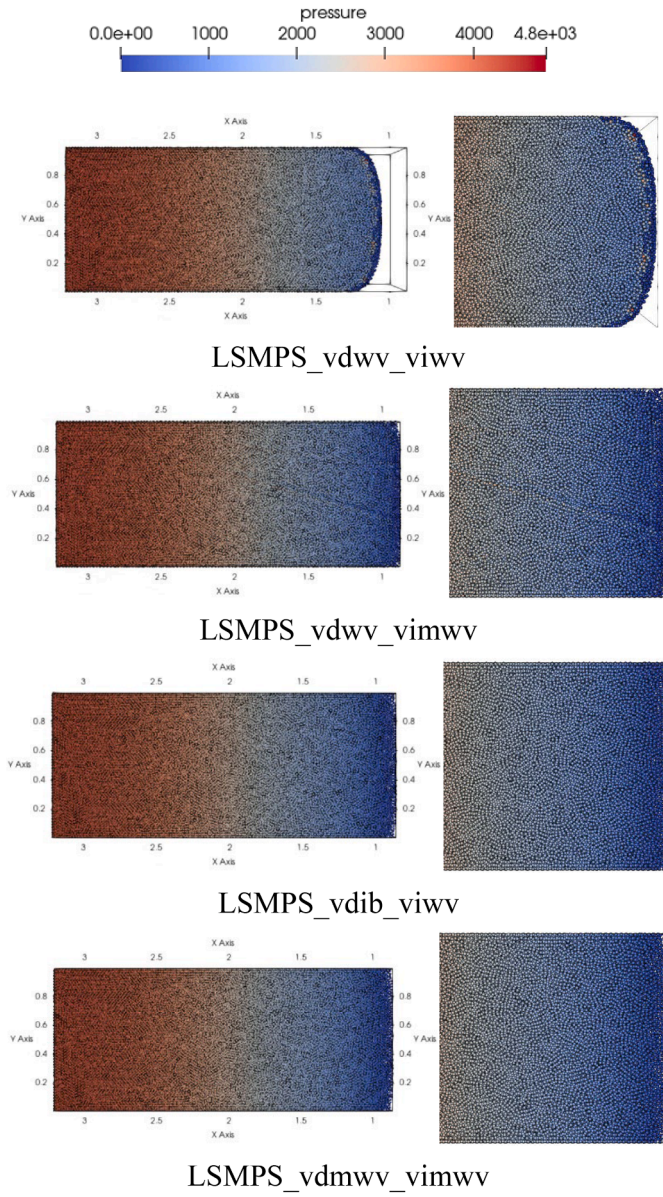


Fig. 2. Particle distribution together with pressure field of the dam at time = 0.4 s (bottom view).

$$\langle \nabla p \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (p_j - p_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \\ \mathbf{L}_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (50)$$

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j - \mathbf{u}_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \\ \mathbf{L}_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (51)$$

It is evident that the presence of a limited number of neighbor particles in proximity to the free surface can lead to a state of instability if the ninth-order matrix is inverted. To enhance stability, the first-order ($p = 1$) method is adopted. Furthermore, the incorporation of type-B, which accounts for the target particle itself, has the potential to further enhance the stability. Thus type-B ($p = 1$) is adopted for both free-surface and sub-free-surface particles:

$$\langle \nabla p \rangle_i = \sum_j \left\{ \frac{w_{ij}}{r_s} (p_j) \left(\begin{bmatrix} \tilde{\mathbf{L}}_{2,p=1} \\ \tilde{\mathbf{L}}_{3,p=1} \\ \tilde{\mathbf{L}}_{4,p=1} \end{bmatrix} \hat{\mathbf{p}}_{ij,p=1}^T \right) \right\} \quad (52)$$

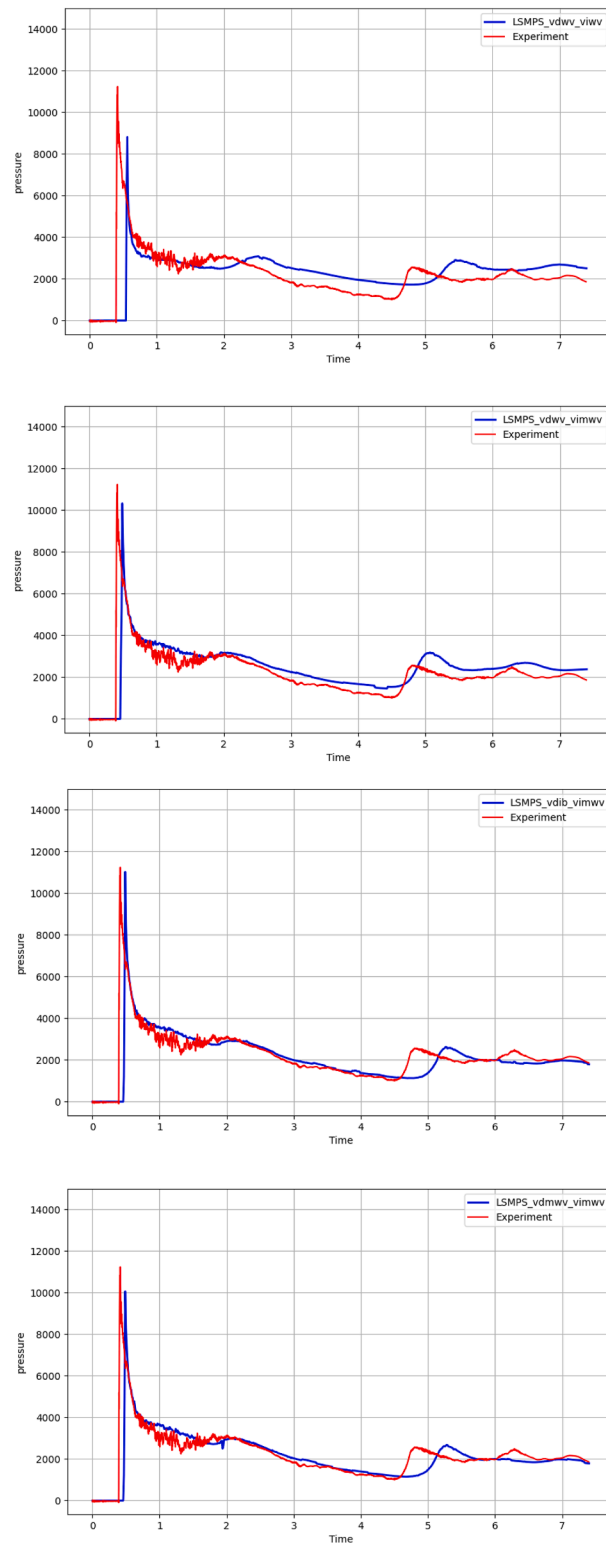


Fig. 3. Pressure profile at the probe position of the three-dimensional dambreak simulation.

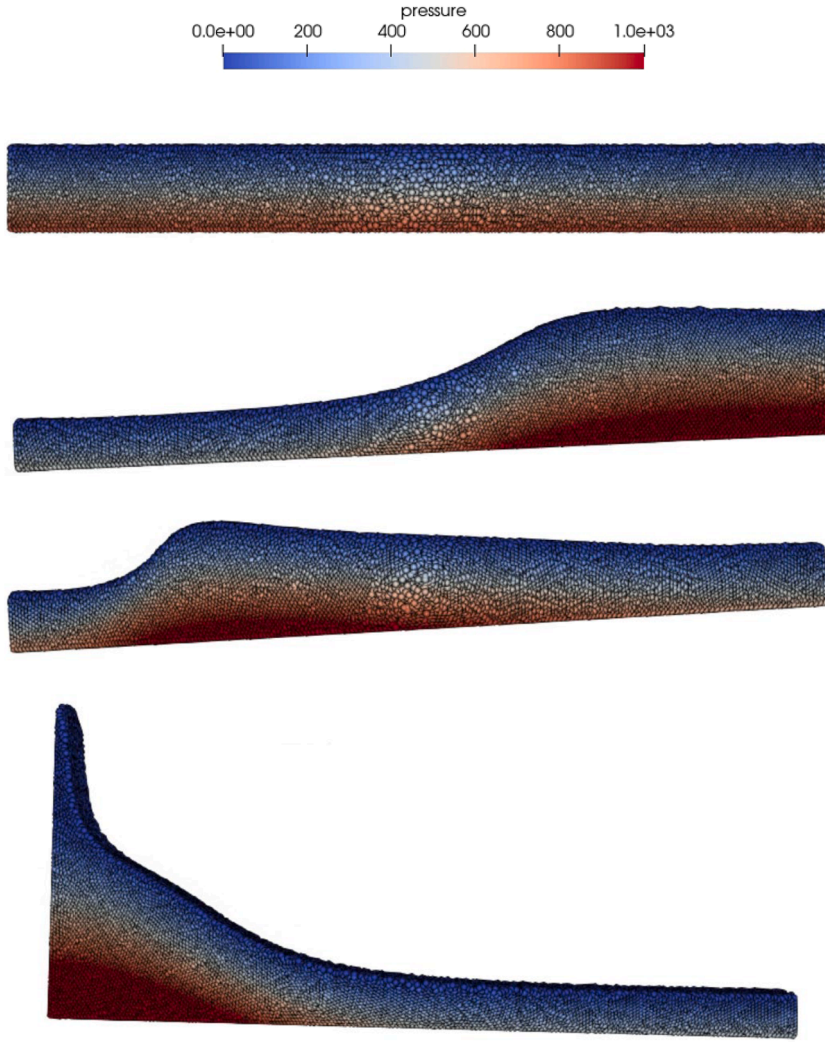


Fig. 4. Particle distribution together with pressure field at time =0.0 s, 2.0 s, 4.0 s and 6.0 s in the sloshing problem.

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \sum_j \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j) \left(\begin{bmatrix} \tilde{\mathbf{L}}_{2,p=1} \\ \tilde{\mathbf{L}}_{3,p=1} \\ \tilde{\mathbf{L}}_{4,p=1} \end{bmatrix} \hat{\mathbf{p}}_{ij,p=1}^T \right) \right\} \quad (53)$$

It should be noted that in the velocity divergence calculation, the proposed velocity wall boundary model (Eq. (45)) is incorporated.

After particle shifting, particle position is shifted from $\mathbf{r}_i$ to $\tilde{\mathbf{r}}_i$. particle velocity at the shifted position is interpolated by type-B LSMPS. Particle velocity at new position after shifting $\mathbf{u}(\tilde{\mathbf{r}}_i)$ is treated as an unknown value. Type-B ($p = 2$) LSMPS is used for internal particles

$$\mathbf{u}(\tilde{\mathbf{r}}_i) = \sum_j \left\{ w_{ij} \left(\tilde{\mathbf{L}}_1 \tilde{\mathbf{p}}_{ij}^T \right) \mathbf{u}(\mathbf{r}_j) \right\} \quad (54)$$

Type-B ($p = 1$) LSMPS is used for free-surface and sub-free-surface particles:

$$\mathbf{u}(\tilde{\mathbf{r}}_i) = \sum_j \left\{ w_{ij} \left(\tilde{\mathbf{L}}_{1,p=1} \tilde{\mathbf{p}}_{ij,p=1}^T \right) \mathbf{u}(\mathbf{r}_j) \right\} \quad (55)$$

3.3. Stability of type-A and type-B LSMPS

LSMPS is required to calculate the matrix inversion. However, in the vicinity of the free surface, the matrix property is suboptimal, which can compromise the precision of the inversion calculation.

Usually, a regularization term is added to improve the stability [21,22,24].

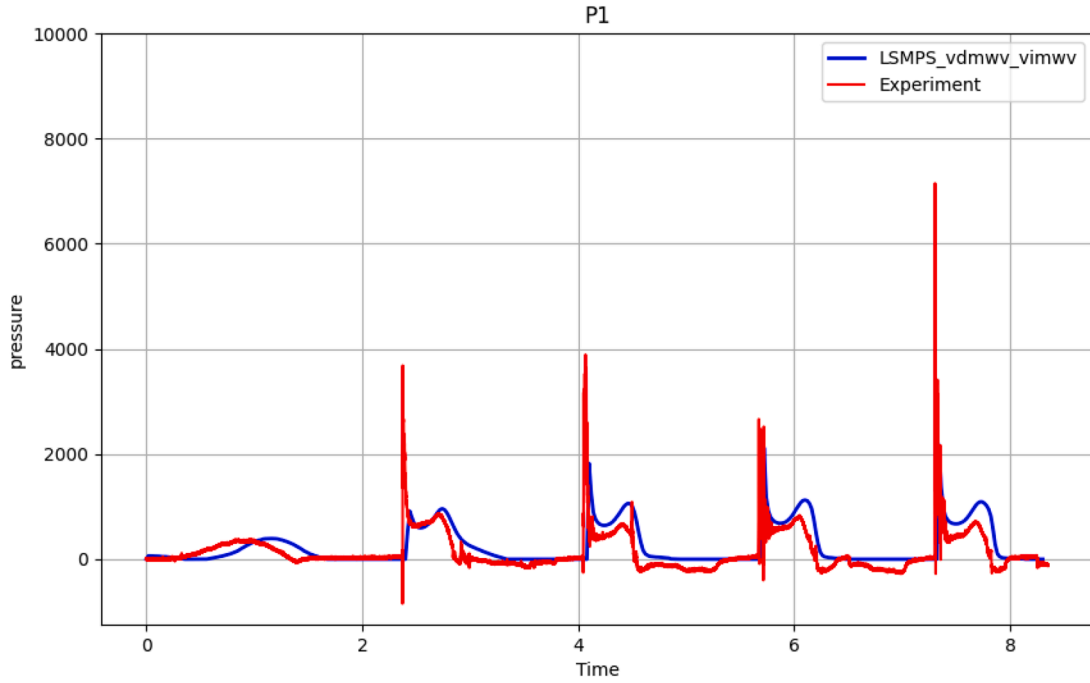


Fig. 5. Pressure profile at the sensor position in the sloshing problem.

For type-A LSMPS:

$$\Lambda \phi_{r=r_i} = \mathbf{H}_s (\mathbf{M}_i + \lambda \mathbf{R}_i)^{-1} \mathbf{b}_i \quad (56)$$

$$\mathbf{R}_i = \text{diag}(1, 1, 1, 1, 1, 1, 1, 1, 1) \quad (57)$$

For Type-B LSMPS:

$$\tilde{\Lambda} \phi_{r=r_i} = \tilde{\mathbf{H}}_s (\tilde{\mathbf{M}}_i + \lambda \tilde{\mathbf{R}}_i)^{-1} \tilde{\mathbf{b}}_i \quad (58)$$

$$\tilde{\mathbf{R}}_i = \text{diag}(1, 1, 1, 1, 1, 1, 1, 1, 1) \quad (59)$$

$$\tilde{\mathbf{b}}_i = \sum_{j \neq i} w_{ij} \tilde{\mathbf{p}}_{ij} \phi_j + \lambda \mathbf{\Pi} \phi_i \quad (60)$$

$$\mathbf{\Pi} = [1, 0, 0, 0, 0, 0, 0, 0, 0]^T \quad (61)$$

In the present study $\lambda=0.001$ is used for sub-free-surface particle while $\lambda = 0.1$ is used for free surface particle.

To ensure the stability of the LSMPS method, the adjustment of particle positions through the particle shifting technique is essential. In this study, the optimized particle shifting (OPS) model, introduced by Khyyer et al. [28], is employed. The OPS model computes the particle shifting vector as follows:

$$\delta \mathbf{r}_i = \begin{cases} -\lambda_{shift} (l_0)^2 (\mathbf{I} - \mathbf{n}_i \otimes \mathbf{n}_i) \nabla D_i & i \in G_{free} \\ -\lambda_{shift} (l_0)^2 \nabla D_i & i \in G_{int} \end{cases}, \quad (62)$$

where λ_{shift} is a constant number, D is the concentration defined as $D_i = \sum_j V_j w_{ij}$ and $\mathbf{n}_i$ is the free surface unit normal vector. λ_{shift} is an adjusted parameter and fixed to 0.01 in this study.

4. Numerical tests

This section presents the results of several numerical tests. In the following tests, numerical simulations were performed using LSMPS_vdwv_viwv, LSMPS_vdwv_vimwv, LSMPS_vdib_vimwv and LSMPS_vdmwv_vimwv. The differences of these methods are explained using second-order ($p = 2$) type-A and type-B LSMPS. Velocity divergence of the provisional velocity was calculated using the provisional wall velocity in vdwv, while modified provisional wall velocity was used in vdmwv.

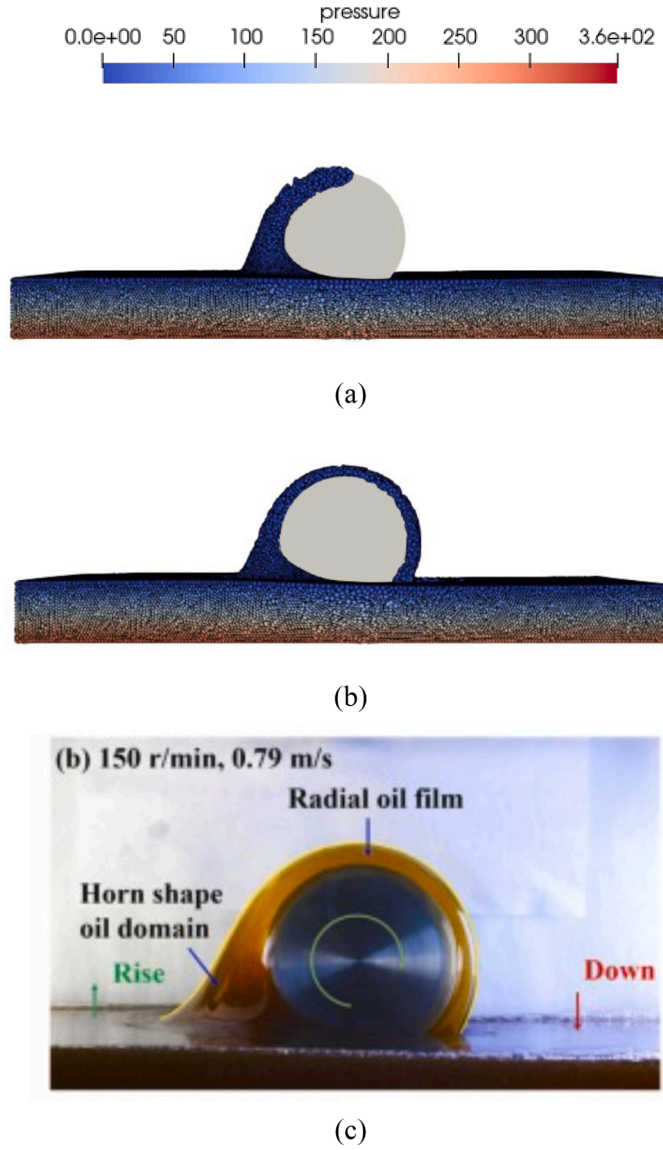


Fig. 6. Oil distribution in the simulation at time = 0.2 s (a) and 0.4 s (b) and oil distribution at steady state in the experiment [31] (c) at rotating speed = 150 rpm.

$$vdwv : \langle \nabla \cdot \mathbf{u}^* \rangle_i = \sum_{j \neq i, j \in \text{fluid}} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j^* - \mathbf{u}_i^*) \left(\begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} + \sum_{j \neq i, j \in \text{wall}} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j^* - \mathbf{u}_i^*) \left(\begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (63)$$

$$\mathbf{u}_j^* = \mathbf{u}_{\text{wall}}^n + \Delta t \mathbf{g} \quad (64)$$

vdmwv:

$$\langle \nabla \cdot \mathbf{u}^* \rangle_i = \sum_{j \neq i, j \in \text{fluid}} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j^* - \mathbf{u}_i^*) \left(\begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} + \sum_{j \neq i, j \in \text{wall}} \left\{ \frac{w_{ij}}{r_s} (\tilde{\mathbf{u}}_j^* - \mathbf{u}_i^*) \left(\begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (65)$$

$$\tilde{\mathbf{u}}_j^* = \mathbf{u}_i^* - (\mathbf{n}_{\text{wall}}^{n+1T} \cdot \mathbf{u}_i^*) \mathbf{n}_{\text{wall}}^{n+1} + (\mathbf{n}_{\text{wall}}^{n+1T} \cdot (\mathbf{u}_{\text{wall}}^n + \Delta t \mathbf{g})) \mathbf{n}_{\text{wall}}^{n+1} \quad (66)$$

The velocity divergence was calculated exactly according to the impermeable boundary condition constrained by Eq. (44) following [19] in vdiib.

Velocity interpolation was calculated using wall velocity in vimv, while velocity interpolation was calculated using modified wall

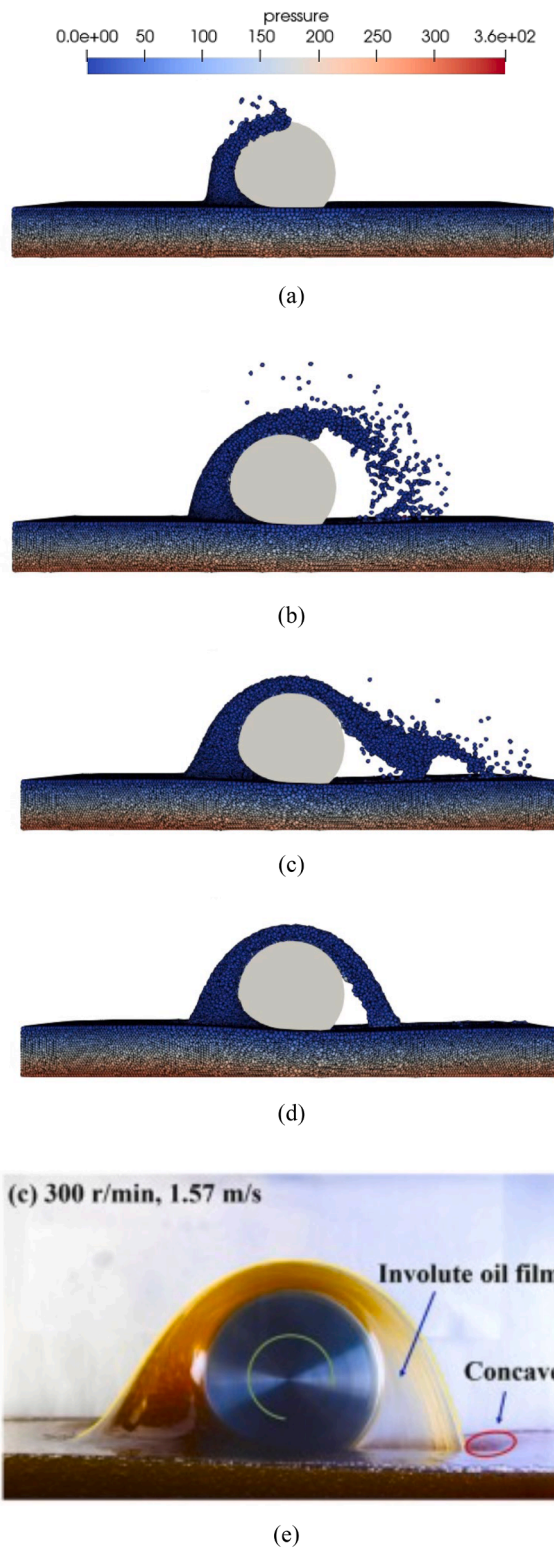


Fig. 7. Oil distribution in the simulation at time = 0.1 s, 0.2 s, 0.3 s and 0.4 s and oil distribution at steady state in the experiment [31] (c) at rotating speed = 300 rpm.

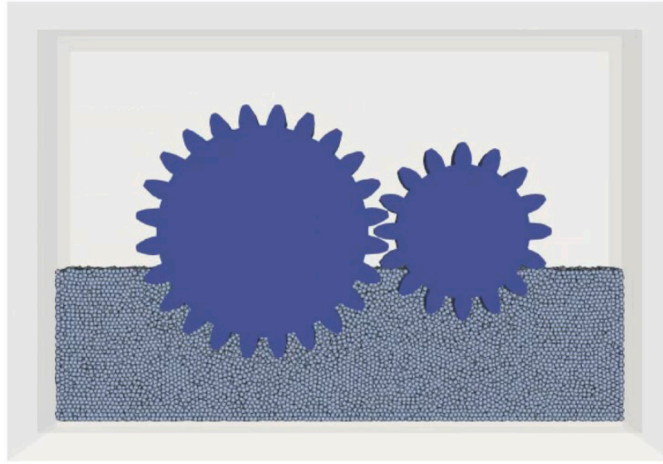


Fig. 8. Initial geometry in the gearbox simulation.

velocity in vimwv.

$$vimv : u(\tilde{r}_i) = \sum_{j \in fluid} \{w_{ij} (\tilde{L}_1 \tilde{p}_{ij}^T) u(r_j)\} + \sum_{j \in wall} \{w_{ij} (\tilde{L}_1 \tilde{p}_{ij}^T) u(r_j)\} \quad (66a)$$

$$vimwv : u(\tilde{r}_i) = \sum_{j \in fluid} \{w_{ij} (\tilde{L}_1 \tilde{p}_{ij}^T) u(r_j)\} + \sum_{j \in wall} \{w_{ij} (\tilde{L}_1 \tilde{p}_{ij}^T) \tilde{u}(r_j)\} \quad (67)$$

$$\tilde{\mathbf{u}}_j = \mathbf{u}_i - \left(\mathbf{n}_{wall}^{n+1 T} \cdot \mathbf{u}_i \right) \mathbf{n}_{wall}^{n+1} + \left(\mathbf{n}_{wall}^{n+1 T} \cdot \mathbf{u}_{wall}^n \right) \mathbf{n}_{wall}^{n+1} \quad (68)$$

It is important to note that the modified wall velocity in Eq. (68) differs from that in Eq. (66). The former equation utilises the provisional wall velocity following the prediction step, whereas the latter employs the wall velocity at the new time step.

4.1. Dam break with obstacle

3D dam break flow against a cuboid obstacle is studied to verify the stabilized LSMPS. The experimental result used for comparison is from [29]. In this simulation case, the height of the water column is 0.55 m, while the length is 1.228 m, and the width is 1.0 m. The distance between front and back vertical walls is 3.221 m. The cuboid obstacle is located at the centerline of dry horizontal bed with respect to the width, and has the length 0.161 m, the width 0.403 m and the height 0.161 m. The distances of cuboid obstacle to the front and the back vertical walls are 0.664 m and 2.396 m, respectively. Pressure at the probe position (0.8245, 0.5, 0.021) is recorded and compared with the experiment.

The distribution of particles, in conjunction with the pressure field of the dam at time = 0.4 s, is illustrated in Figs. 1 and 2. A comparison of the pressure profile at the probe position is presented in Fig. 3. A lateral perspective reveals that the water front in LSMPS_vdvw_viwv exhibits a reduced velocity compared to other simulation results. This disparity can be attributed to the utilization of wall velocity in the calculation of fluid velocity after particle shifting in LSMPS_vdvw_viwv, a process that significantly impacts fluid velocity in proximity to the bottom wall. As depicted in Fig. 2, the pressure noise observed in LSMPS_vdvw_viwv and LSMPS_vdvw_vimwv is attributable to the use of wall velocity in the calculation of velocity divergence in PPE. In the absence of modification, the velocity gradient along the tangential direction is substantial, giving rise to pressure noise. However, the application of an impermeable wall condition (LSMPS_vdib_vimwv) or a modified wall velocity (LSMPS_vdmwv_vimwv) results in a reduction of pressure noise.

As illustrated in Fig. 3, the initial pressure peak manifests at a subsequent stage in LSMPS_vdvw_viwv, a finding that aligns with the observations presented in Fig. 1. The pressure profile derived from other methods exhibits a comparable trajectory to the experimental result. A minor discrepancy is evident in the timing of the pressure peak. This can be attributed to the velocity interpolation following particle shifting. It is imperative to note that velocity interpolation is a prerequisite for ensuring the stability of calculations, albeit resulting in the introduction of numerical viscosity.

The simulation results presented in this section demonstrate that the modified wall velocity model employed in vdmwv and vimwv models significantly enhances the simulation outcomes in comparison to vdvw and viwv. The simulation result by vdmwv is also comparable with the result by vdib. The lsmws_vdmwv_vimwv model is utilised in the following numerical tests.

4.2. Sloshing under forced rolling motion

In this section, the stabilized LSMPS (LSMPS_vdmwv_vimwv) is verified by the sloshing problem under force rolling motion in [30].

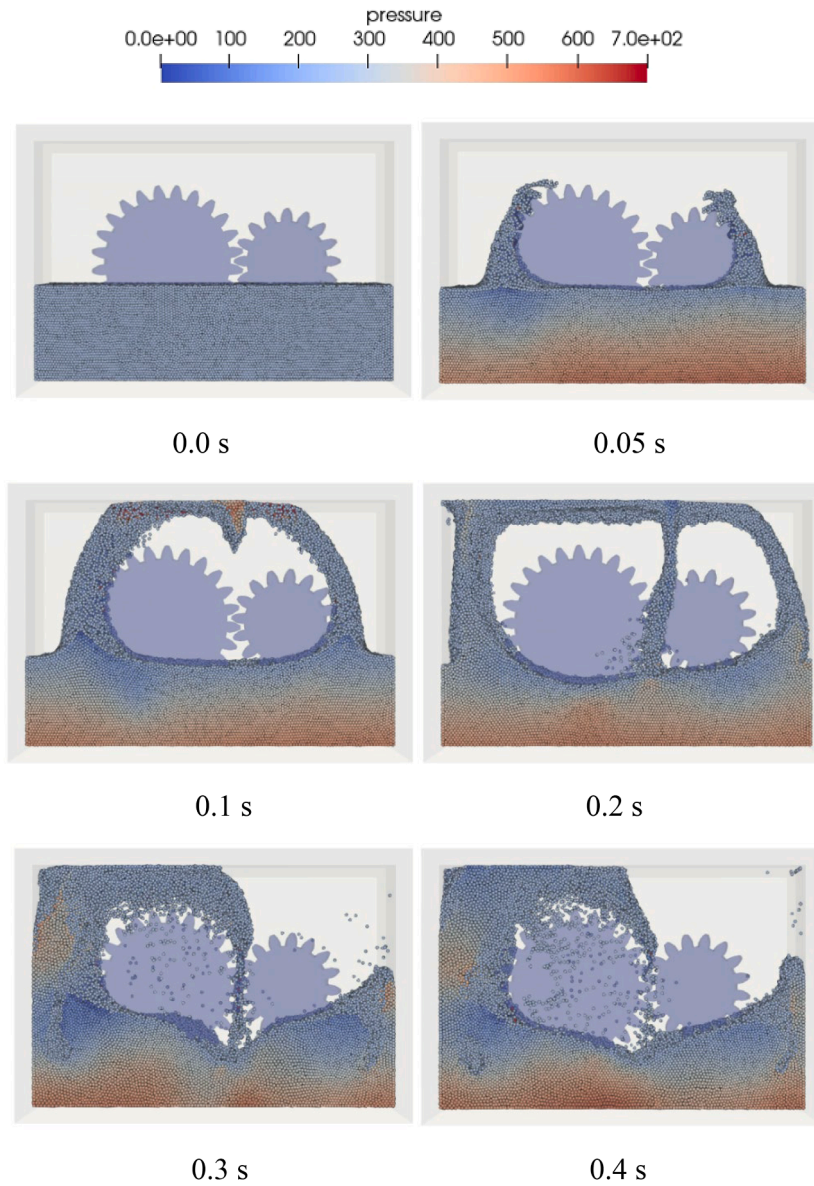


Fig. 9. Oil distribution in the gearbox simulation (rotating speed = 1444 rpm, oil temperature is 40 °C).

In this problem, the size of the tank is 0.9 m×0.58 m×0.05 m. The tank is excited via an electrical motor with a sinusoidal type rotating motion. Water depth is 0.093 m. The corresponding first sloshing period and excitation periods are both 1.92 s. A pressure sensor is placed on the tank wall, at the initial height of the free surface, on the left side. Particle length in the simulation is 0.004 m.

As illustrated in Fig. 4, the distribution of particles and the pressure field at specific times (0.0 s, 2.0 s, 4.0 s and 6.0 s) are presented. It is evident that the free surface undergoes significant deformation under excitation, which is distinctly captured by the movement of particles. Fig. 5 compares the pressure at the sensor position. The five pressure peaks observed in the simulation are analogous to the experimental results.

4.3. Rotating disk

In this section, the rotating disk in [31] is simulated to verify the stabilized LSMPS method. In this problem, a disk with diameter = 10 cm is partly immersed in the oil and rotating under different rotating speed. The immersing length is 2 cm. The height of the oil pool is 40 mm. The density and viscosity of oil is 800 kg/m³ and 0.0015 m²/s, respectively.

Two cases with different rotating speed are calculated. In the first case, the rotational speed is set at 150 rpm (rpm). Fig. 6 illustrates the distribution of oil at specific intervals of 0.2 s and 0.4 s. The oil is observed to flow along the periphery of the disk and subsequently

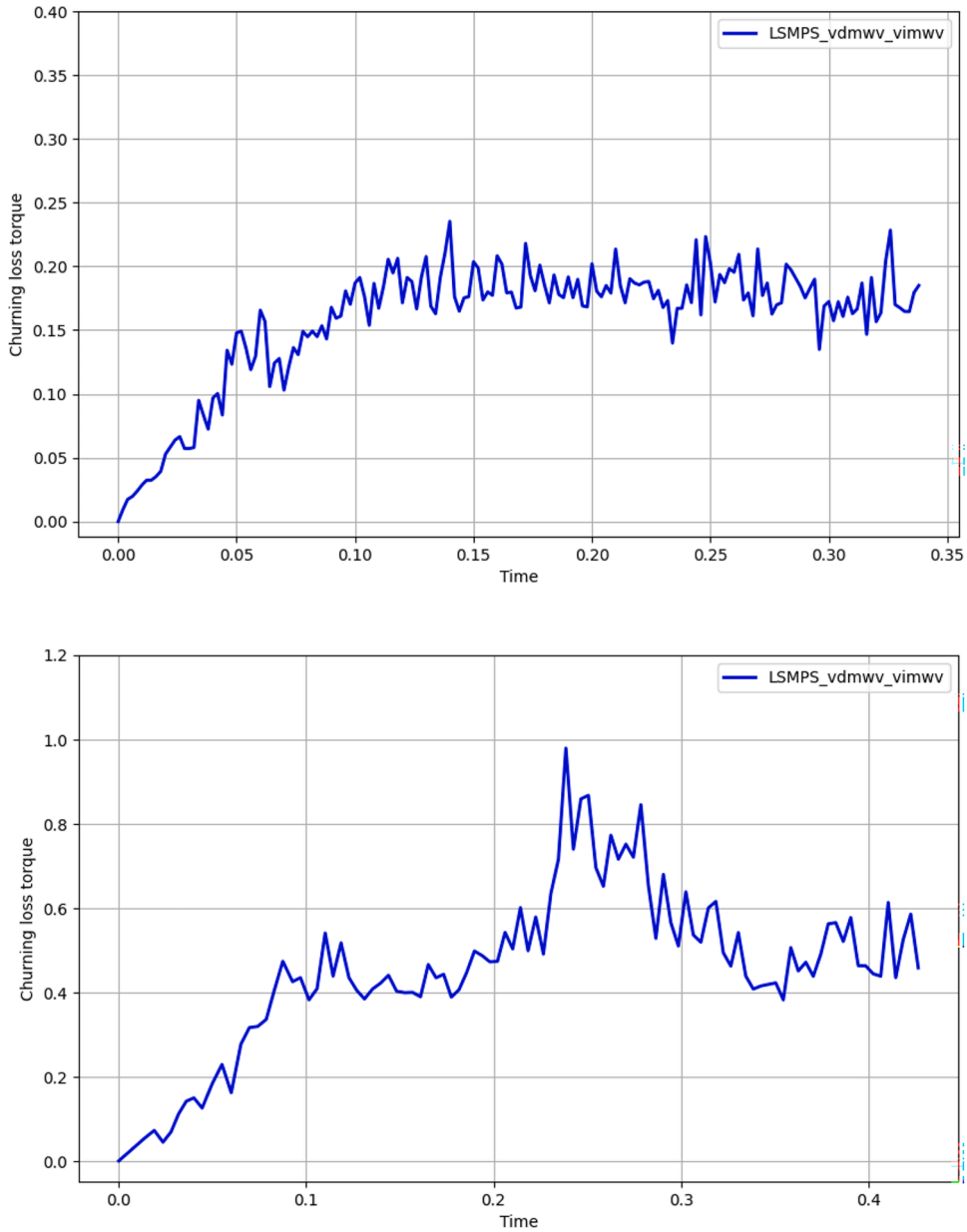


Fig. 10. Churning loss torque at rotating speed = 1444 rpm and 3474 rpm.

descend into the designated oil pool. The observed flow pattern of the oil demonstrates a high degree of congruence with the experimental outcome.

In the second case, the rotating speed is 300 rpm. Fig. 7 shows oil distribution at times of 0.1 s, 0.2 s, 0.3 s and 0.4 s. Due to the higher rotating speed in comparison with the first case, the oil film does not coat the surface of the disk, and separation can be observed. Finally, the flow pattern at steady state is in very good agreement with the experimental result.

4.4. Gearbox

In this section, the gearbox problem in [32] is simulated. Initial geometry is shown in Fig. 8. The size of the oil pool is 210 mm×70 mm×56 mm, oil fill level is 20 mm below the middle axis of the gear. Pinion and the wheel are rotating under different rotating speed.

In this study, two cases from [32] are simulated. The first case compares the oil distribution. The oil temperature is set at 40 °C, the rotating speed at 1444 rpm, the kinematic viscosity of the oil at 90 mm²/s, and the density of the oil at 884.1 kg/m³. The oil

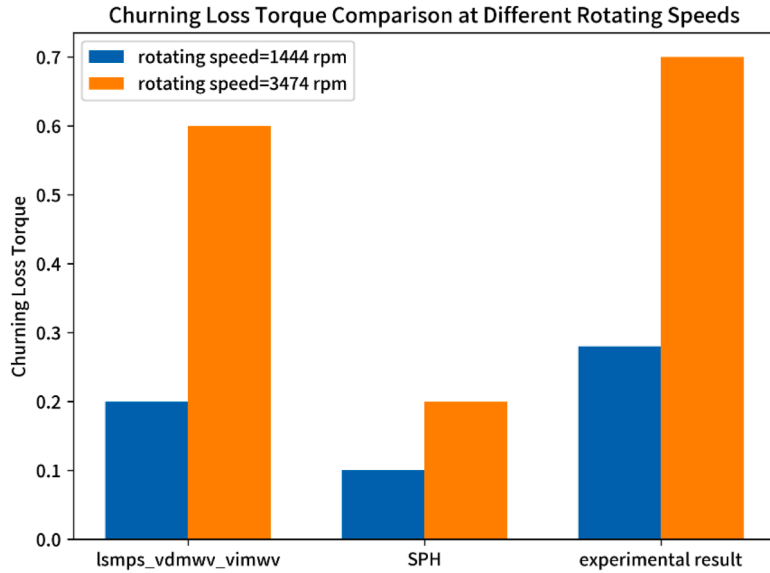


Fig. 11. Comparison of the churning loss torque with SPH result and experiment result in [32].

distribution at different time instants is shown in Fig. 9. The oil flows along the gearbox and reaches the top side of the tank. The oil that has accumulated at the top of the gearbox rolls up by the pinion and subsequently collides with the oil that has accumulated at the top of the wheel, causing it to fall down through the gap between the pinion and the wheel. The oil continues to be accumulated at the top of the wheel, while the oil level at the pinion side drops down. The simulated result by SPH and experimental result can be found in [32]. The simulation result, which was obtained by using the developed stabilized LSMPS method, agrees better with the experimental result compared to the SPH result in [29].

In the second case, the churning loss torque is compared. The torque is calculated by

$$\mathbf{T} = \sum_i \mathbf{r}_i \times (\mathbf{F}_{i,pressure} + \mathbf{F}_{i,viscosity}) \quad (69)$$

Here $\mathbf{T}$ is the total churning loss torque of fluid particle on the gears, $\mathbf{r}_i$ is the distance vector from the center of the gear to the fluid particle i . $\mathbf{F}_{i,pressure}$ and $\mathbf{F}_{i,viscosity}$ is pressure force and viscosity force of particle i acting on the gears.

The oil temperature is set at 90 °C, and the kinematic viscosity of the oil is 15 mm²/s. The density of the oil is 884.1 kg/m³. Fig. 10 illustrates the calculated profiles of churning loss torque at two distinct rotational speeds: 1444 rpm and 3474 rpm. The average churning loss torque after 0.1 s is compared with the simulation results obtained via the SPH method and experimental data from [32]. The results of this comparison are presented in Fig. 11. In the experimental analysis, the churning loss torque was approximated to be 0.3 Nm and 0.7 Nm respectively at the two speeds. Although simulated values may somewhat underestimate the experimental torque, the simulation results generated by the LSMPS_vdmwv_vimwv method show good agreement with the experimental data. This indicates the effectiveness of the LSMPS_vdmwv_vimwv approach in predicting churning loss torque. A sudden rise in torque around 0.22 s can be observed when the rotating speed is 3474 rpm, as illustrated in Fig. 12. This is due to the oil being rolled up by the gears and then hitting them as it falls, thereby producing extra torque.

5. Conclusion

In this study, a stabilized LSMPS was developed by mixing the type A and type B LSMPS for free surface with moving wall. A modified wall velocity boundary condition is proposed to allow accurate moving wall boundary computation. The stability of the type A and type B schemes has been improved by adding positive values to the diagonal element of the least squares matrix.

Dam break with obstacle, sloshing and rotating disk are simulated to verify the stabilized LSMPS. Simulation results show the advantage of using modified wall velocity in velocity interpolation and velocity divergence calculation. The simulation results by the developed method show good agreement with the reference result. Gearbox is also studied by the stabilized LSMPS method. The simulating results agree well with the experimental result in terms of oil distribution and churning power loss.

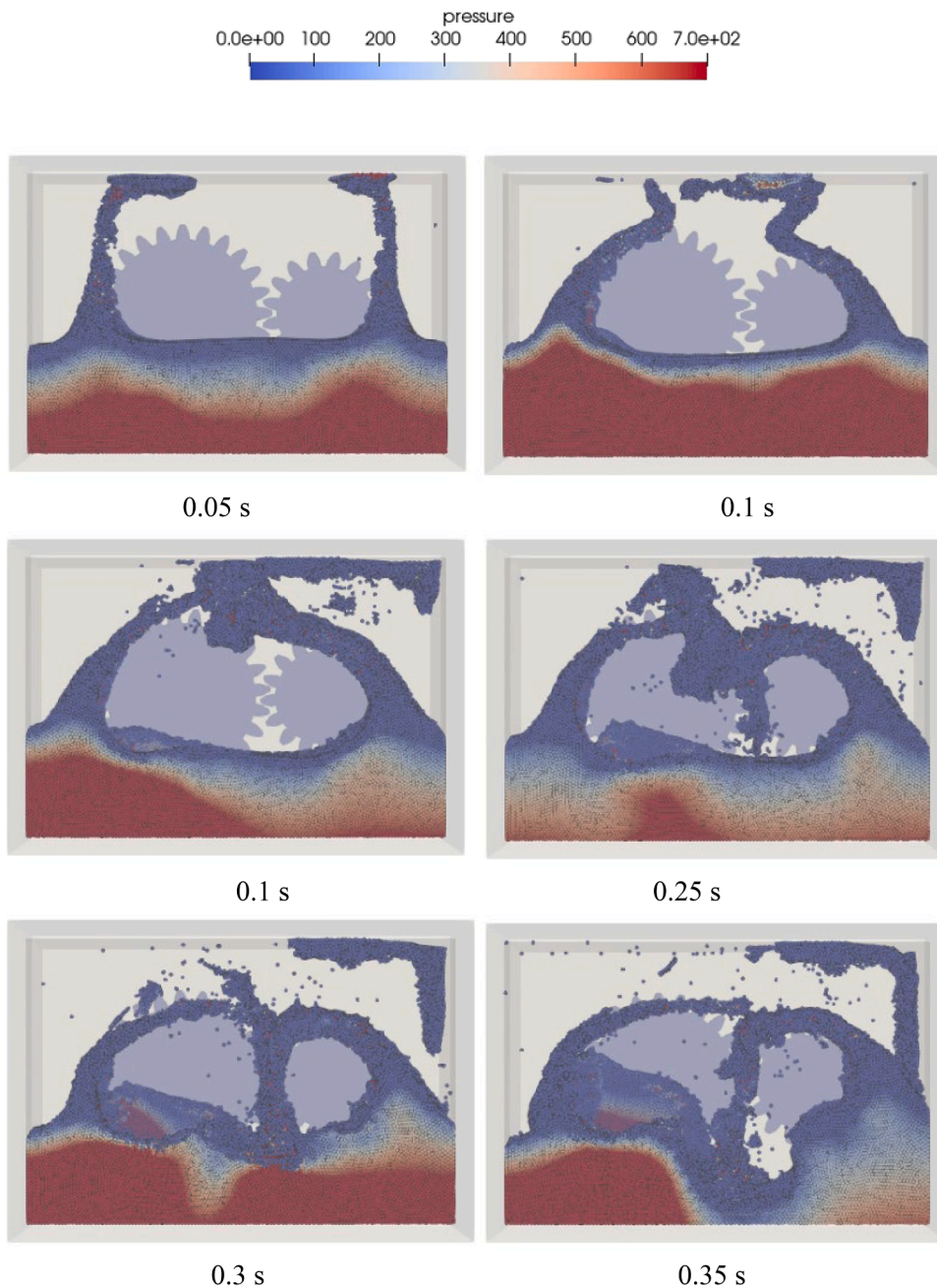


Fig. 12. Oil distribution at different time instants at the rotating speed = 3474 rpm.

CRediT authorship contribution statement

Xiaoxing Liu: Writing – original draft, Validation, Methodology, Conceptualization. **Siwei He:** Writing – review & editing, Methodology, Conceptualization. **Wei Huang:** Validation, Methodology, Conceptualization. **Xi Wang:** Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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